

Chapter 1

Introduction

Neutron stars (NSs) are believed to be some of the densest objects in the Universe, coming second only to black holes. Their interiors host supranuclear densities and comparatively low¹ temperatures—a phase of matter under active research, which is simultaneously motivated by a need to describe astrophysical observations of NSs and an incomplete physical description of the strong force at supranuclear densities. Because the extreme combination of gravitational forces and matter is not reproducible in terrestrial laboratories, theoretical attempts reliant on physics-agnostic quantum chromodynamics (QCD) (e.g. R. Somasundaram et al., 2023) have been used to constrain the NS equation of state (EoS). Complementarily, because the star is an observable astrophysical object, properties such as the speed of sound (e.g. T. Kojo, 2020), mass, radius, and tidal deformability (e.g. A. Hegade K. R. et al., 2026) have also been used to constrain the EoS of neutron stars, and by extension, the behavior of dense QCD matter in extreme conditions.

Although it is difficult to constrain NS observables solely from electromagnetic (EM) detections, given their comparatively feeble luminosity in most nonextremal EM spectra (A. Y. Potekhin et al., 2020), tidal deformability can also be constrained by gravitational waves (GWs) emitted during binary neutron star (BNS) inspirals (e.g. J. Faber & R. Frederic, 2012), illustrated (T. Dietrich et al., 2021) in Fig. (1.1). Explicitly, tidal deformability is typically expressed in terms of dimensionless quantities called the Love numbers², first introduced by A. Love (1909) to describe the Earth’s approximate susceptibility to the deformation due to tidal fields (e.g., by the Sun and the Moon), *a linear and adiabatic response to the deformation-inducing field*. This concept of deformability has

¹The low temperature criteria here describe the temperature in the nuclear setting. Given that the Fermi energy of the constituent particles are many orders of magnitude higher than the kinetic thermal energy, we can treat the pressure relation as temperature-independent (e.g. A. Y. Potekhin, 2010), where the latter is taken to approach 0.

²In this thesis, we take Love numbers to be time-independent quantities. It is not uncommon to describe *dynamic* Love numbers, which describe dynamic tidal deformation, but we will describe such quantities differently.

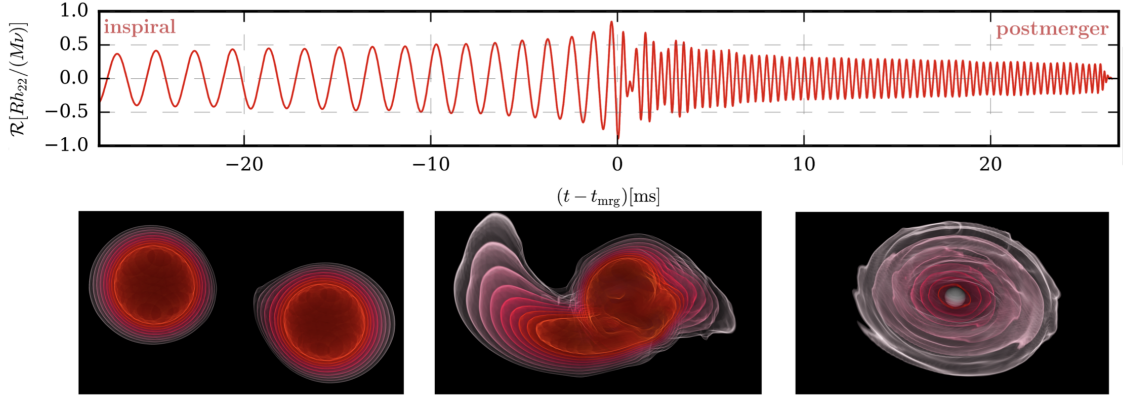


Figure 1.1: The numerical relativity simulation [figure directly sourced from T. Dietrich et al. (2021)] results of a BNS merger and the resulting gravitational waveform emitted about the merger time (*top panel*), as well as the rest-mass density evolution (*bottom panels*) during the late inspiral, merger, and ringdown/postmerger.

since been generalized to describe the deformation as a response to the tidal field within the general relativistic framework, and the relevant quantities are described in §5. Specifically, the deformation of a neutron star, which has long been predicted to occur due to a compact binary companion³ (e.g. a NS or black hole, BH), can also be described by a set of Love numbers (T. Binnington & E. Poisson, 2009). Such tidal quantities have been thought to be extractable from the waveform of the GW emitted from the compact binary.

Both predictions⁴ were empirically confirmed by gravitational waves resulting from a (likely) BNS merger and its resulting kilonova, which were detected on August 17, 2017 (GW170817). The observation was made by the Advanced LIGO detectors (Hanford, Livingston) and spatially constrained by a Virgo GW-detector (Pisa) (B. P. Abbott et al., 2017). The pre-merger binary masses have been constrained by B. Abbott et al. (2019) (in Table III) to a 90% confidence interval⁵ of $m_1 = 1.46^{+0.12}_{-0.10} M_\odot$, $m_2 = 1.27^{+0.09}_{-0.09} M_\odot$. The event itself is associated with the fourth loudest GW signal observed to date (see GW250114, GW230814, GW231226 respectively; The LIGO Scientific Collaboration and the Virgo Collaboration and the KAGRA Collaboration, 2025; A. G. Abac et al., 2025). It is worth mentioning that GW190425 is also widely believed to have been produced by a BNS merger (e.g. Y. Qin et al., 2024), although B. P. Abbott et al. (2020) mention that either companion cannot be ruled out as a black hole due to the merger’s unusually high source-frame

³This thesis focuses specifically on BNSs as such GW sources, as EM counterparts of BH-NS binaries have yet to be detected and more relevantly, large-mass BHs would merge quickly with the NS, allowing little time for the former to tidally disrupt the latter (H. Koehn et al., 2024).

⁴The deformation of the neutron star due to a compact binary companion, as well as the encoding of the deformation in the waveform of the GW emitted.

⁵See Table III for assumptions regarding the priors.

chirp mass and total mass.

Figure (1.1) demonstrates how GWs emitted from the BNS merger vary with the type of tidal interaction that dominates. Throughout the inspiral, the companion-induced multipole moments tidally deform both neutron stars, prompting changes in binding energy that increase the rate of energy emission in the form of gravitational waves (L. Blanchet, 2014). This implies an increasing loss rate in the Keplerian energies of both neutron stars, leading to their inspiral and eventual collision (E. E. Flanagan & T. Hinderer, 2008), which can be seen in the growing amplitude of the GW throughout the inspiral.

Love numbers capture the approximately time-independent tidal response theory further out before the merger, during the early part of the late, quasi-circular inspiral. In this case, the response theory can be approximately captured in a related coefficient Λ_2 , the (electric-type, quadrupolar) *static tidal deformability* (T. Hinderer, 2008; K. Chatziioannou, 2020), which enters the gravitational wave-phase at 5 post-Newtonian (PN) order (see §2.2 & §3.2) and has been used to constrain the parameter space of the neutron star EoS (e.g. B. P. Abbott et al., 2018). However, the quasi-static tidal response approximation becomes increasingly inaccurate closer to the merger time, as the tidal effects become dynamical, i.e., dependent on the frequency of the orbit (e.g. A. Hegade K. R. et al., 2024a). The tidal response function of the NS becomes complex in this regime, where the real part is called the *conservative tidal response* and the imaginary part the *dissipative tidal response* (e.g. A. Hegade K. R. et al., 2024b, §3.2), where the former reduces to the static tidal deformability in the zero frequency limit (E. E. Flanagan & T. Hinderer, 2008) and the latter, when limited to the low frequency regime, is approximately the (electric-type, quadrupolar) constant Ξ_2 (see §3.2).

Typically, the nonnegligible contributions to the dynamical tide regime⁶ are described by the coupling (e.g. A. R. Counsell et al., 2025) between the tidal force and the f, g, i -modes. It is well-established that in most quasi-circular binary systems, the strongest contribution is due to the f -mode (fundamental mode) of the NS, which describes the pressure-driven global density oscillations of the star and typically occurs on the order of $\mathcal{O}(2\text{kHz})$ (S. B. Kokkotas, 1999). It is worth mentioning that a significant number of works ignore all other contributions aside from the f -mode contributions, approximating the NS tidal response as a harmonic oscillator with the fundamental mode equal to the f -mode frequency of the star (e.g. J. Steinhoff et al., 2016; T. Hinderer et al., 2016; P. Schmidt & T. Hinderer, 2019, §4.1), which have been used to constrain the f -mode frequencies of NSs (e.g. T. Hinderer et al., 2016).

⁶Given the low-frequency, constant approximation of the dissipative tidal deformability, we only consider frequency-dependent effects on the conservative tidal deformability.

However, certain binary setups can observe non-negligible effects from excitations in the lower frequency modes, such as the g -mode. The g -mode (gravity mode) describes the buoyancy-driven forces resulting from NS thermal and composition gradients. In addition, discontinuities in such gradients produce a phase transition from confined quark matter (confinement in hadrons) to deconfined quark matter, yielding potentially non-negligible resonance modes during tidal deformation (A. R. Counsell et al., 2025), inducing i -mode (interfacial modes, also known as discontinuity g -modes) resonances. Typically, the oscillations occur on the order of $\mathcal{O}(100\text{Hz})$ (A. Reisenegger & P. Goldreich, 1992) and $\mathcal{O}(600\text{Hz})$ (e.g. A. R. Counsell et al., 2025; A. Hegade K. R. et al., 2026) respectively.

Although the time-independent tidal interactions can sufficiently constrain the static deformabilities of NSs using current GW data, the frequency range of the dynamical contributions is currently insufficiently resolved and will have to rely on increased detector sensitivity to model and distinguish between different time-dependent tidal responses of NSs (e.g. Cosmic Explorer; N. Williams et al., 2022; P. Baral et al., 2023). In anticipation of such high-sensitivity detectors, physically-motivated studies have moved past the (static) Love number approach to tides, whether in the exploration of nonlinearity in the tidal deformability of the neutron star (e.g. T. Pitre & E. Poisson, 2025; P. Pani et al., 2025) or the influence of dynamical tides on the GWs emitted from BNS mergers, where one forgoes the tide-adiabaticity assumption (e.g. T. Hinderer et al., 2016). Extensions of BNS tidal dynamics to the frequency-dependent regime have been made in various ways, one of them being a *phenomenological* approach (A. Hegade K. R. et al., 2026) that restricts to Newtonian-like mode approximations in effective one-body models (e.g. J. Steinhoff et al., 2016), which are then adjusted against numerical relativity simulations (A. Hegade K. R. et al., 2026). However, such an approach makes it difficult to quantify and characterize the approximation errors. Another method relies on asymptotic expansions in the PN restriction, which allows for error reduction by including higher order PN terms (e.g. A. Hegade K. R. et al., 2024a).

While such studies focus on the *linear* tidal response, the highly nonlinear nature of the merger has prompted some groups to study the impact of nonlinear tidal interactions on the dynamics of BNS systems. H. Yu et al. (2023) and T. Pitre & E. Poisson (2025) motivated such investigations using a Newtonian and general relativistic treatment of the BNS tidal dynamics to show that nonlinear contributions to the tidal interactions are nonnegligible in accurately modeling the effect of dynamical tides on BNSs. Namely, the latter showed that the PN expanded tidal response can also be described in terms of a (frequency-domain) harmonic oscillator, and that the (quadratic) Love number p_2 contribution lowered the frequency parameter of the tidal oscillation model by up to

15% in polytropic stellar models. Given that the resonance is reached earlier in the inspiral in the nonlinear scheme, the importance of nonlinear contributions to the accuracy of BNS tidal effects is a direct consequence of T. Pitre & E. Poisson (2025).

However, the nonlinear post-Newtonian contributions to the tidal dynamics of BNSs have yet to be studied in the GW waveform phase picture. Addressing this gap using post-Newtonian theory is our primary goal. We begin our analysis in §2, which briefly reconstructs the many-body orbital mechanics in the Newtonian framework, then extends to the PN framework whilst covering fundamental aspects in PN theory, which then motivates the derivation of the quadrupole formula for gravitational waves. §3 serves an introduction to Newtonian tidal dynamics and the assumptions made in the static and dynamic tidal regimes, as well as an examination of tidal effects using linear response theory. §4 follows the derivation (E. E. Flanagan & T. Hinderer, 2008) of static tidal deformation on the Newtonian BNS orbital dynamics and provides a means of calculating its effects on the phase of the resulting GW waveform. §5 follows J. L. Ripley et al. (2023) in the examination of dynamic tides on the Newtonian BNS orbital dynamics in the quasicircular approximation using perturbation theory, which is then generalized to nonlinear perturbations without the linear response framework presented in §5.1. The derived tidal phase correction is compared to the results derived by P. Pani et al. (2025) using effective field theory and quadratic perturbation theory. §6 derives the linear contributions to the binary dynamics in the Post-Newtonian framework. §7 summarizes our findings and their potential applications in gravitational wave astronomy and nuclear physics. We also discuss the linear and nonlinear contributions in the PN framework, to be addressed in future work.